



NPEC CV & Robotics





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CLIENT

02

CHALLENGE

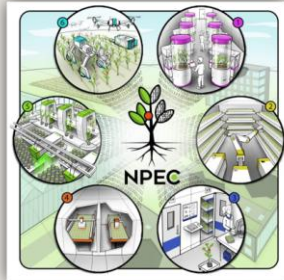
03

SOLUTION

04

RESULTS





ABOUT NPEC

NPEC is a research facility operated by two Dutch universities. The center provides advanced automated systems for plant phenotyping -- measuring observable plant characteristics under different environmental and genetic conditions.

PHENOTYPE

01

GENOME

Changes in genome are fundamental

02

ENVIRONMENT

Even with the same DNA environment influence more

03

INTERACTION

The result is clear only after intercourse of DNA and Environment

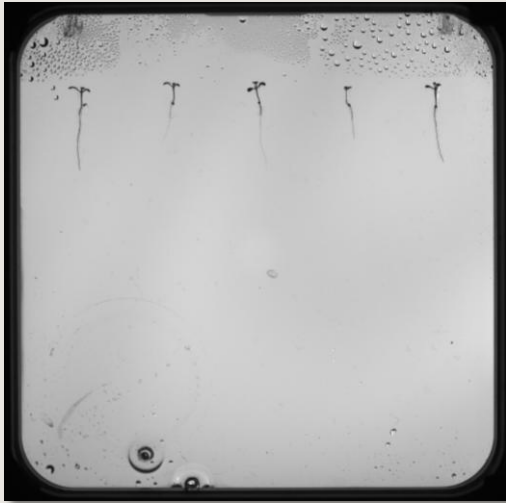
04

ITERATION

To understand the nature of plants we need a lot of cases



SECTION 1 : THE CHALLENGE



COMPUTER VISION + ROBOTICS

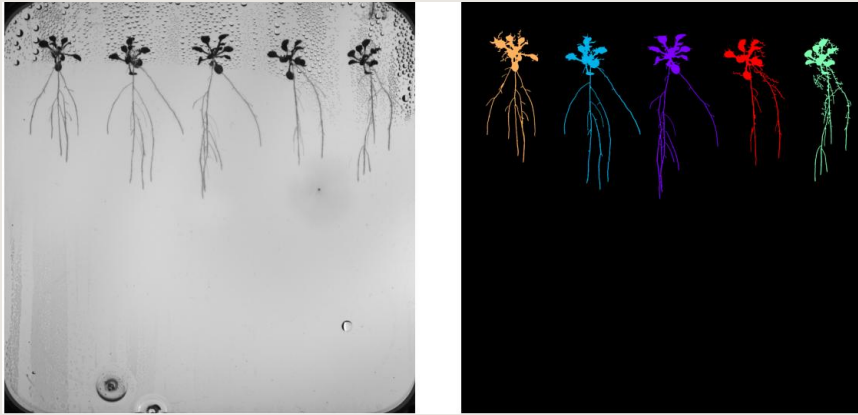
OT-2

❖ *Arabidopsis thaliana*

- Precision
- Scale
- Speed
- Reliability
- Cost



WHY?



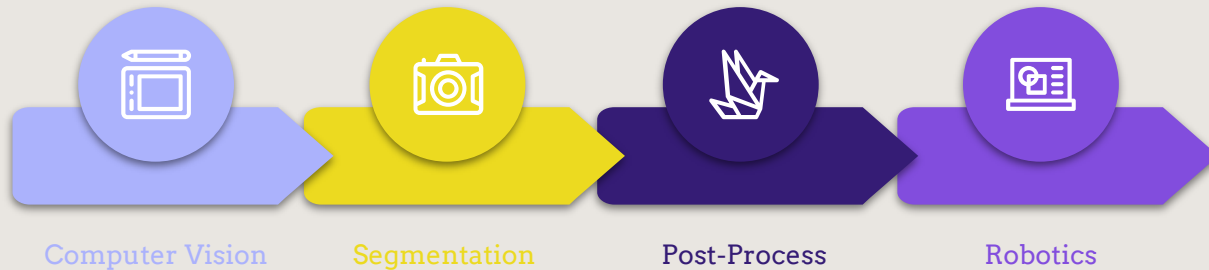
Manual – Long, Inefficient

Disadvantages:

- Time
- Noise
- Cost
- Unreliable



SECTION 2: PROPOSED SOLUTION



Assumptions:

- Expected 5 plants per plate
- Fixed Petri dish location in simulation
- Only one camera position
- Roots are mostly vertical



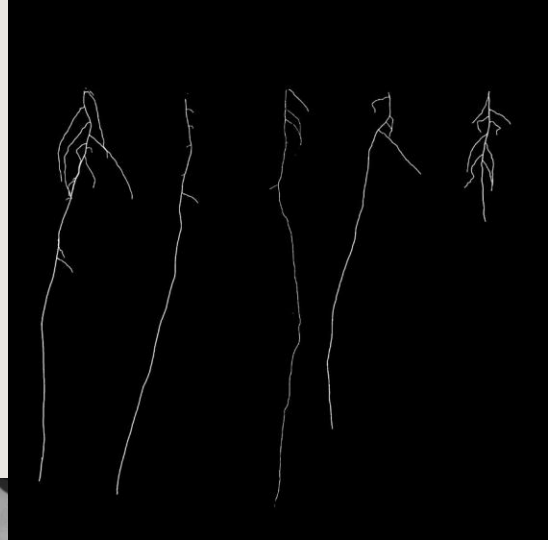
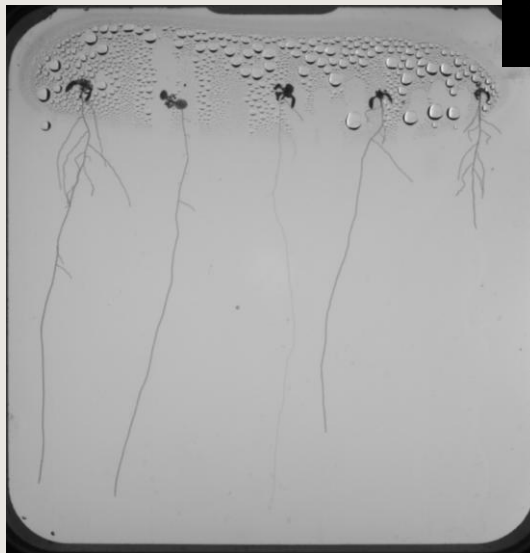
COMPUTER VISION I

U-NET

U-NET + ResNet34

Attributes:

- *Encoder:* ResNet34
- *Decoder:* Standart U-Net (4)
- *Input:* Grayscale (256x256)
- *Output:* Binary Segmentation
- *Loss:* DICE (0.5x) + Focal (0.5x)
- *Optimizer:* AdamW (1e-4)
- *Augmentation:* Elastic Transform

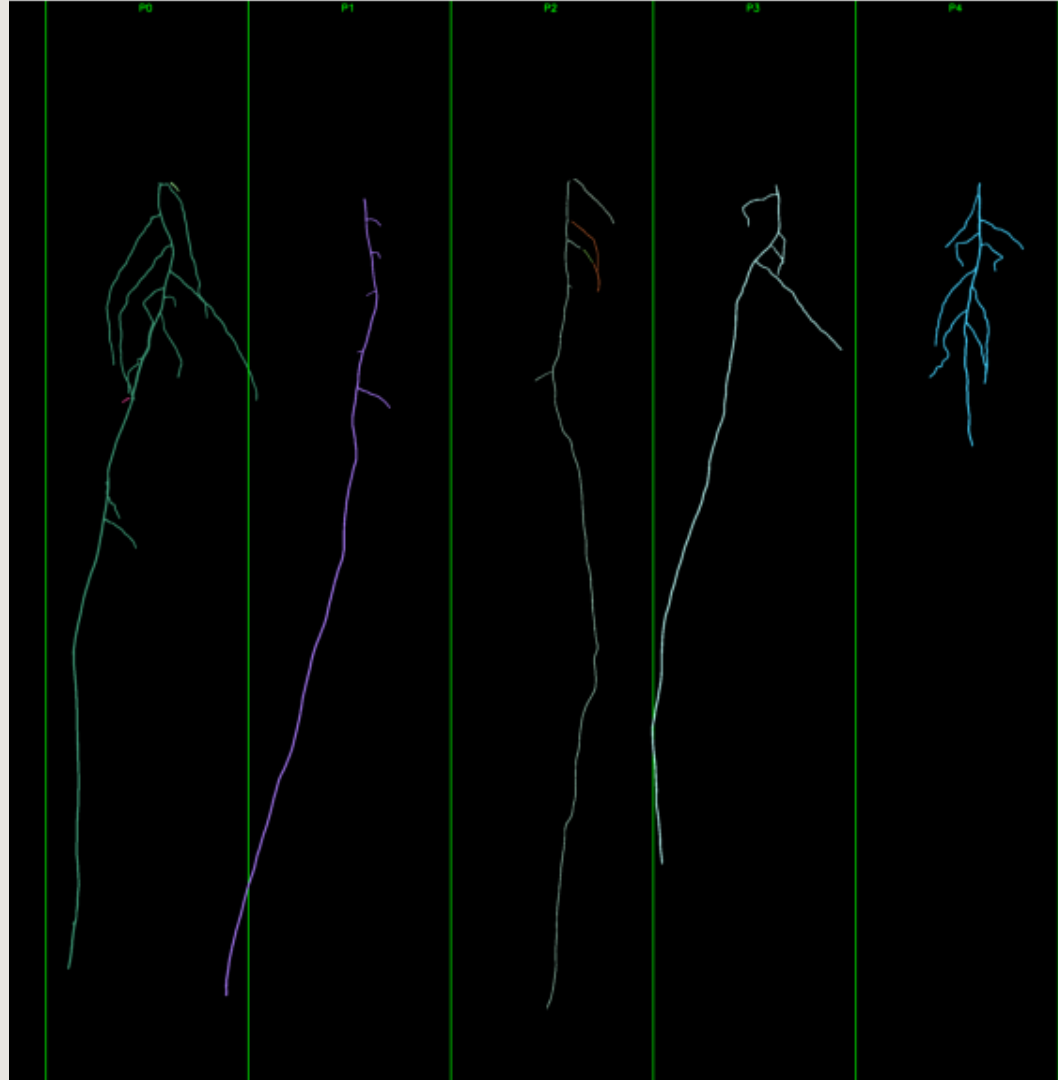


COMPUTER VISION | POST-PROCESS

Segmentation

Attributes:

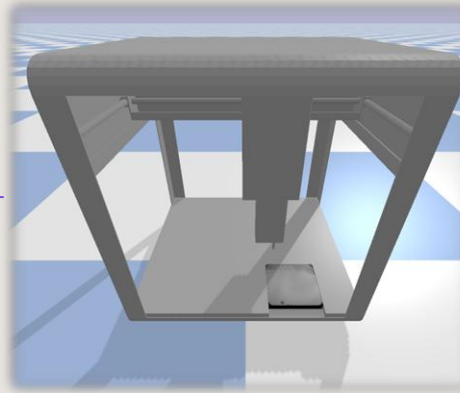
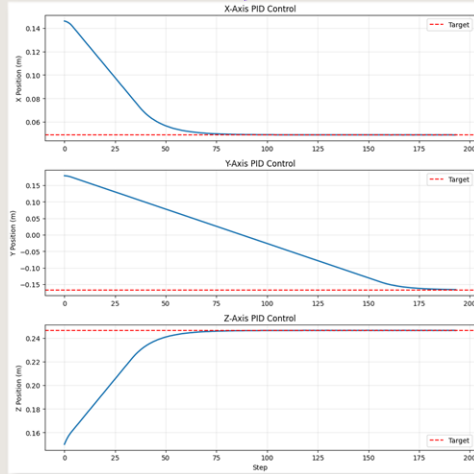
- Expected Plants: x: 350, step: 500
- ROI: Each zone ± 250 px
- Smart Component Assignment
- ROI vs Component overlap
- Scipy Lable function in each region
- Outlier detection (too far)
- H/V Component analysis



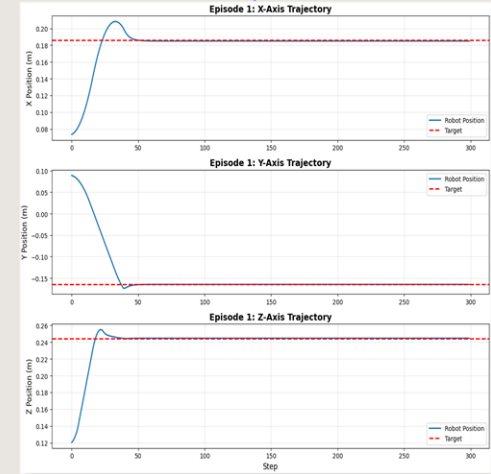
SIMULATION

ROBOTICS

PID CONTROLLER



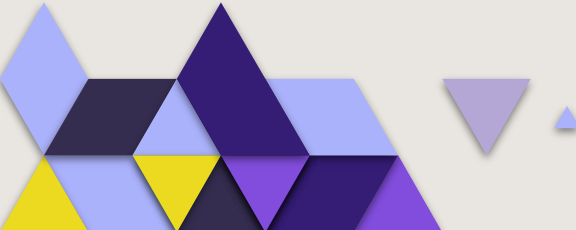
RL CONTROLLER



FINAL ERROR

AVG 0.9 MM

AVG 1.2 MM



SECTION 3: VALUE TO THE CLIENT

Speed
Automatic solution just works at any time stably



Precision
Submillimeter accuracy on average



Scale
Much easier to scale automatic production



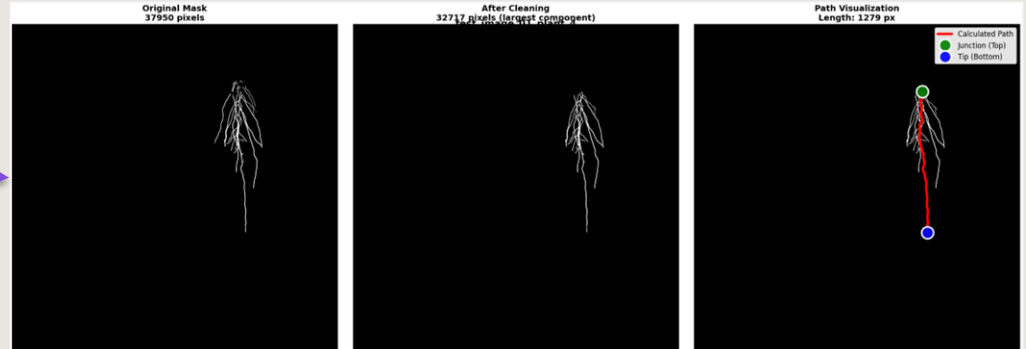
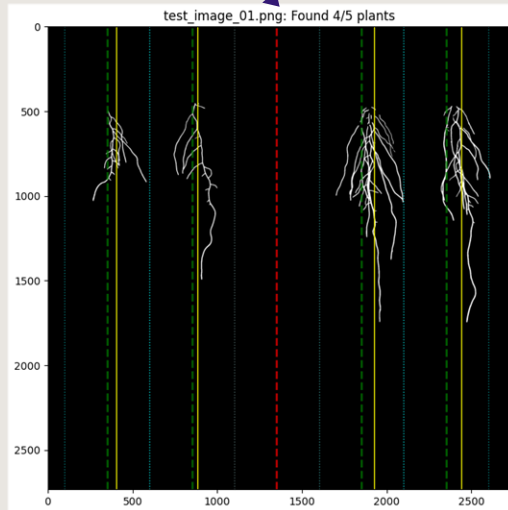
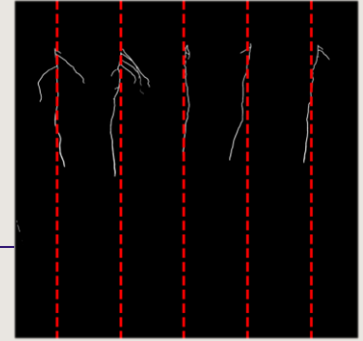
Discovery
More time – More experiments
– More discoveries



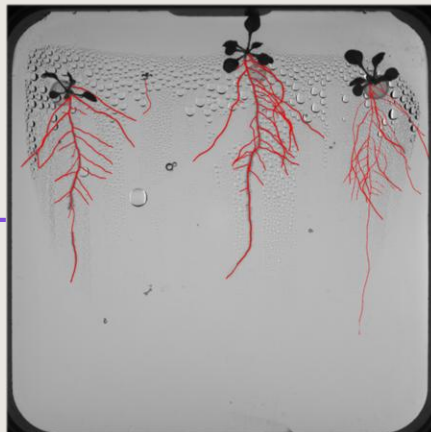
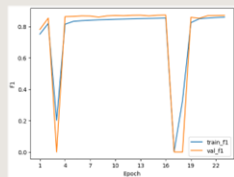
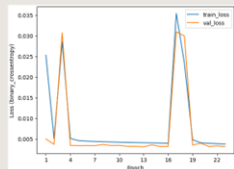
SECTION 4: RESULTS & EVALUATION

INFERENCE + SEGMENTATION

CV Pipeline

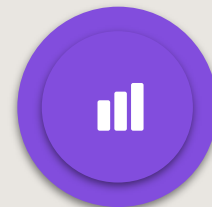


METRICS



VAL LOSS: 0.003

VAL F1 SCORE: 0.87



KAGGLE | SCORES

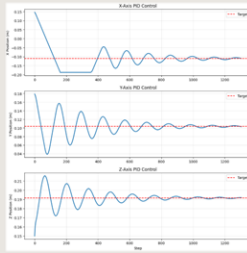
#	△	Team	Members	Score	Entries	Last	Solution
1	▲ 1	Razvan Nica 242715		5.021	27	8d	
2	▲ 11	Alex Krasnoshtanov		5.125	8	4d	
3	▲ 18	Lars Peggeman 241863		5.612	39	9d	
4	▲ 4	Gergő honyák		5.672	25	1d	
5	▲ 4	Abdelrahman Samara 245769		5.747	16	2d	

FINAL sMAPE: 5.125

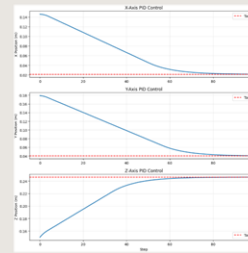
FINAL RANK: 2



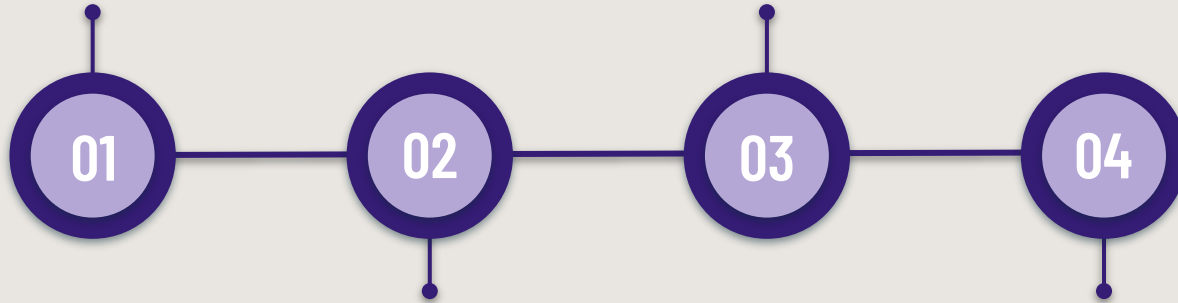
PID TUNING



First Iteration,
Random Parameters



Best Avg Parameters,
Found after 30+ test cases



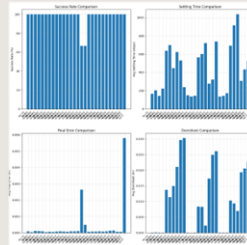
FINAL kP: 20

FINAL kI: 0

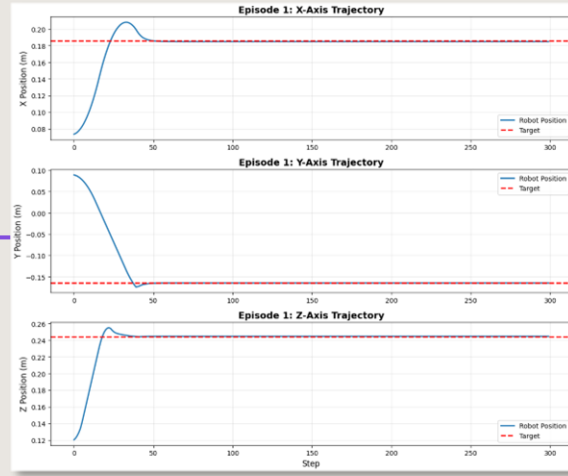
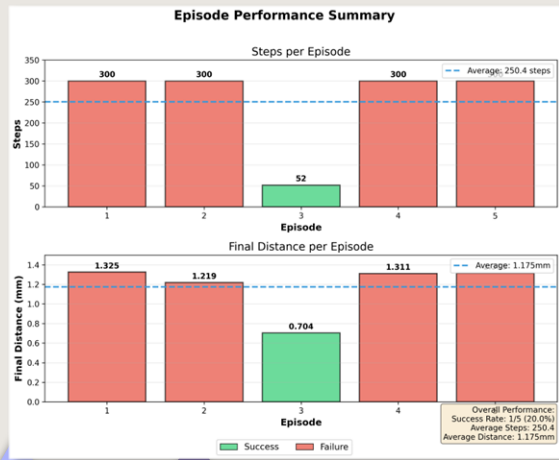
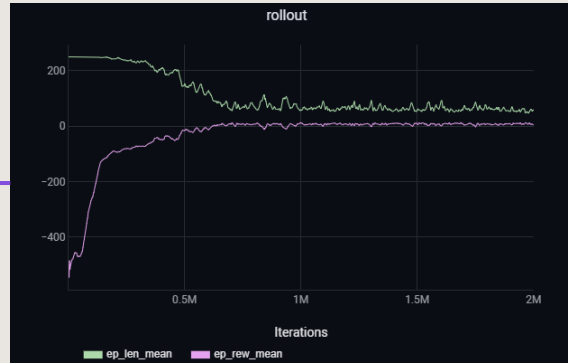
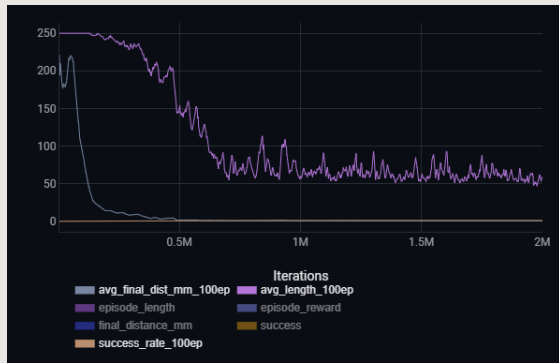
FINAL kD: 1

FINAL AVG ERROR: 0.94mm

Tuning



Future Steps,
Distance based varying parameters,
the test shown that on different
distance, different parameters are
becoming more efficient, so making
them adaptive would make average
use more effective



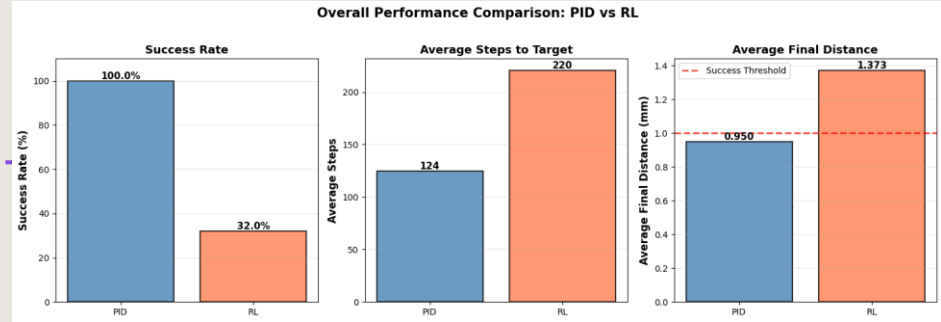
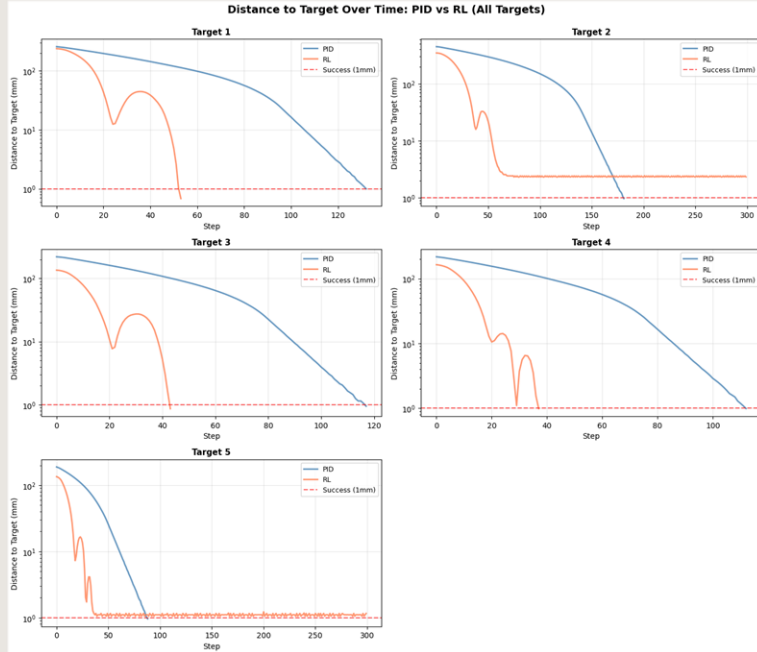
RL Controller

PPO Reinforcement Learning

Attributes:

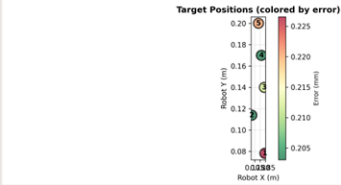
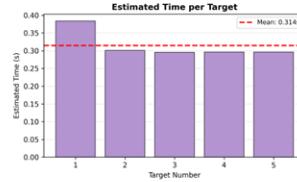
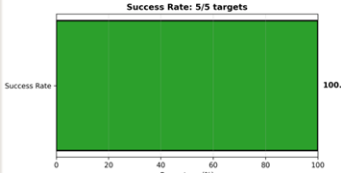
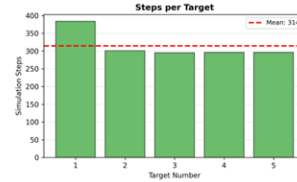
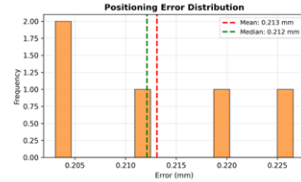
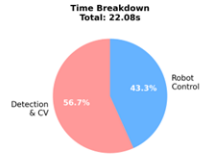
- *Batch Size: 128*
- *Gamma: 0.99*
- *Learning Rate: 0.001*
- *Max Steps Truncate: 250*
- *N_Steps: 2048*
- *Target Threshold: 0.001*
- *Total Timesteps: 2000000*

COMPARISON: PID | RL



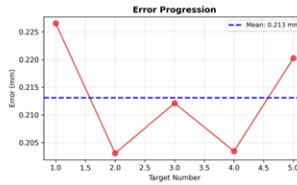
We can see that on a large scale (Second graph is based on 100 runs of PID vs RL on the same target coordinates) PID outperforms the current implementation of PPO Reinforcement Learning algorithm

Robot Dispensing System - Performance Analysis 20260116_021019

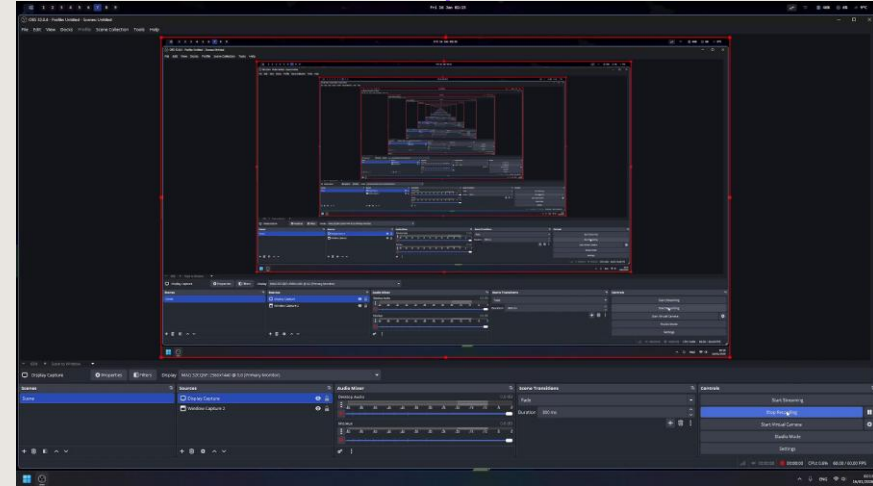


PERFORMANCE SUMMARY

Processing Time:	
• Detection & CV:	12.53s
• Robot Control:	9.55s
• Total Time:	22.08s
• Avg per Target:	1.93s
Accuracy Metrics:	
• Mean Error:	0.213 mm
• Median Error:	0.212 mm
• Max Error:	0.227 mm
• Min Error:	0.203 mm
• Std Dev:	0.009 mm
Efficiency:	
• Total Steps:	1,572
• Mean Steps:	314
• Total Targets:	5
• Success Rate:	100.0%



SYSTEM INTEGRATION



ERROR ANALYSIS | ITERATIONS

1

First Submission
Too much cleaning,
worse result, 12.5
public sMAPE

2

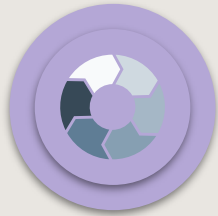
Second-Fourth
Wrong
direction, 20
public sMAPE

3

Final Submission
Success, 2.9
public sMAPE and
understanding

✓ primary_root_lengths (6).csv Complete · 4d ago · One more try	5.125	4.918	✓
✓ primary_root_lengths (6).csv Complete · 4d ago · Without filters	5.411	2.998	✓
✓ primary_root_lengths (5).csv Complete · 5d ago · one more try	8.658	4.965	□
✓ primary_root_lengths (4).csv Complete · 5d ago · one more try	5.411	12.596	□
✓ primary_root_lengths (1).csv Complete · 5d ago · One more time with better algorithm	5.397	12.588	□
✓ primary_root_lengths2.csv Complete · 6d ago · New algorithm	10.212	20.443	□
✓ primary_root_lengths.csv Complete · 7d ago · Test2	8.658	12.554	□

LIMITATIONS | FUTURE



Reinforcement Learning

Not even $<1\text{mm}$ on average

Adaptive PID

Can Process Faster

Better Search Algorithm

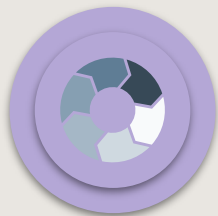
More Accurate Calculations

Crossing Roots

The Classical Problem



LIMITATIONS | FUTURE



Reinforcement Learning

Improve Reward Function

Adaptive PID

If PID still better then RL

Better Search Algorithm

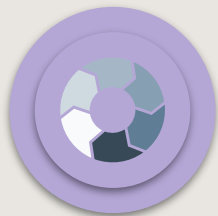
Cover all edge cases

Crossing Roots

Implement Visual Transformers



SUMMARY



Final Solution

Functioning Pipeline

sMAPE, kaggle

Position, Potential

Value

Speed, Quantity, Reliability

Technical

ROI, Roots, Expectetions



The background features a light gray surface with three abstract geometric patterns of triangles in purple, yellow, and blue. One pattern is in the top right corner, another in the bottom right corner, and a third in the bottom left corner. The triangles are arranged in a way that creates a 3D effect, with some triangles appearing to be stacked on top of others. The word "THANKS!" is centered in the middle of the image in a bold, dark gray font.

THANKS!